

# Work, Power & Energy.

- Work - It is a scalar product of force and displacement.

$$W = \vec{F} \cdot \vec{S} = |F| |S| \cos \theta$$

Unit = Nm  $\rightarrow$  Joule.

- When work done is positive then the system gets energy.
- When work done is negative then the system loses energy.

\* For variable force

$$W = \int_{x_1}^{x_2} F dx$$

\* The area of force & displacement gives work done in graph



\* It is not necessary that if displacement is zero, total work done is also zero.

\* For forces in multiple directions

$$W = \int F d\vec{s}$$

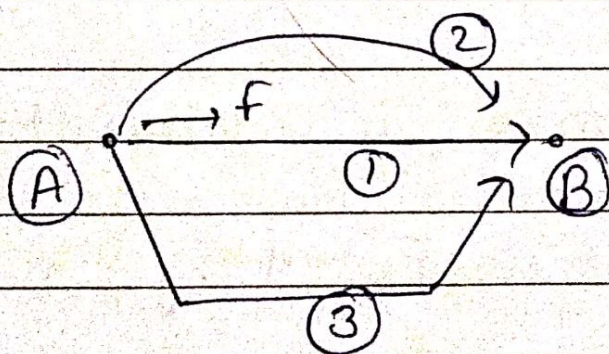
where  $d\vec{s} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ .

\* Conservative force

• The work done by such force

1. does not depend upon ~~the~~ path

2. Depends only & only on initial & final position.



$$W_1 = W_2 = W_3$$

\*) Ex - gravitational force, spring force, electrostatic force.

\* Work done by conservative force in a closed path is always zero.



\* Non conservative forces are forces which are not conservative.

| * Conservative force               | Non Conservative force    |
|------------------------------------|---------------------------|
| 1) $W_I = W_{II}$                  | $W_I \neq W_{II}$         |
| • does <u>not</u> depend upon path | • does depend on path     |
| • Closed path<br>$W = 0$           | Closed path<br>$W \neq 0$ |

(\*) If a Force is conservative

then  $F = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$

Check i)  $\frac{\delta F_x}{\delta y} = \frac{\delta F_y}{\delta x}$

ii)  $\frac{\delta F_x}{\delta z} = \frac{\delta F_z}{\delta x}$

iii)  $\frac{\delta F_y}{\delta z} = \frac{\delta F_z}{\delta y}$



Q. Find whether given force  $x\hat{i} + y\hat{j}$  is conservative or non-conservative.

$$(1) \frac{dF_x}{dy} = \frac{dF_y}{dx} \Rightarrow 0 = 0$$

[  $F_x$  will be constant w.r.t  $y$  ]  
[  $F_y$  will be constant w.r.t  $x$  ]

$$(2) \frac{dF_y}{dz} = \frac{dF_z}{dy} \Rightarrow 0 = 0$$

$$(3) \frac{dF_x}{dz} = \frac{dF_z}{dx} \Rightarrow 0 = 0$$

$\Rightarrow$  Given force is conservative.

## \* Work Energy Theorem.

- Kinetic energy comes into action when body is in motion.
- It depends upon frame of reference.



## • work energy theorem.

⇒ Work done by all the forces on a body = Change in k.e. of the body.

→ internal + external force.

$$W_{\text{by all force}} = K.E._f - K.E._i$$

$$dW_{\text{all forces}} = d(K.E.)$$

## \* Potential Energy

$\Delta U = -$  Work done by conservative force.

$$\Delta U = -W_c$$

$$dU = -dw_c$$



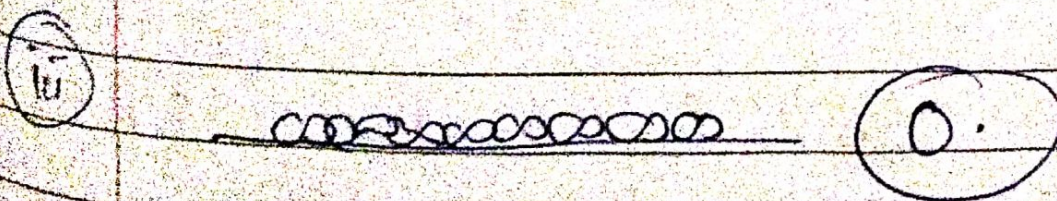
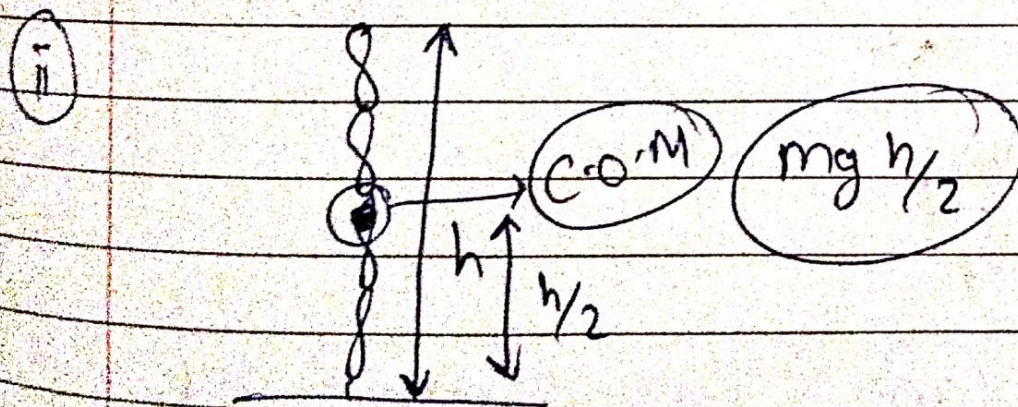
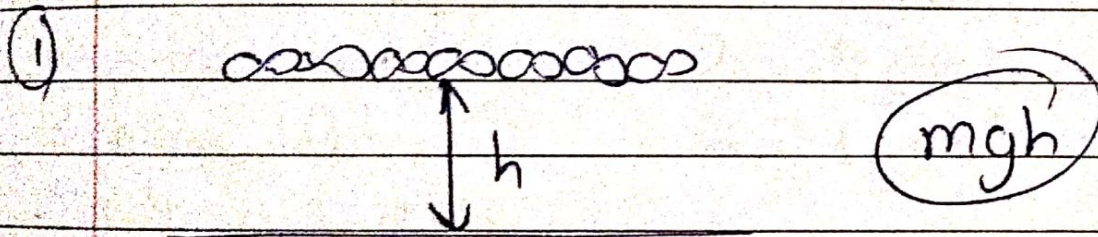
\* Potential energy is not defined for non-conservative force.

$$\int du = - \int dW_c$$

$$\int_{u_i}^{u_f} du = - \int F_c \cdot dx$$

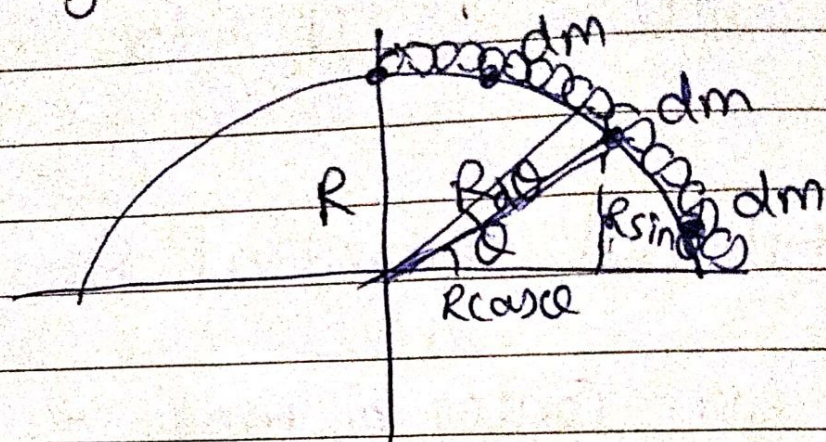
\* Gravitational Potential Energy. ( $mgh$ )

• Check for Center of mass of for  $h$





\* To find Potential Energy of given arrangement of chain.

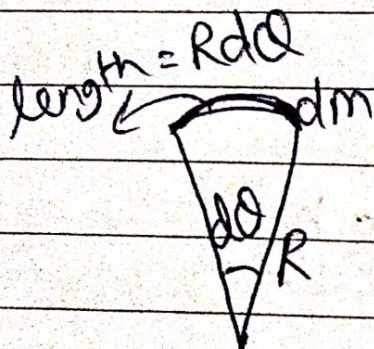


$$dU = dmgh$$

$$\Rightarrow dU = dm g R \sin \theta$$

$$\Rightarrow \int dU = \int dm g R \sin \theta$$

[Here variable is ' $\theta$ ' not  $dm$ , hence make  $\theta$  variable]



Mass  $(m)$  for length  $\frac{\pi R}{2}$

then mass  $dm$  for length  $R d\theta$

$$\Rightarrow \frac{m R d\theta}{\frac{\pi R}{2}} = ?$$

$$dm \Rightarrow \frac{2m d\theta}{\pi}$$



\* Potential Energy is invisible form of energy  
it is imparted to the system, that energy  
is stored as potential energy.

$$\int dU = \int_0^{\pi/2} \frac{2m}{\pi} d\theta g R \sin \theta$$

$$= \frac{2m}{\pi} g R \left[ \sin \theta \right]_0^{\pi/2}$$

$$\Rightarrow \frac{2m}{\pi} g R \left[ -\cos \theta \right]_0^{\pi/2}$$

$$\Rightarrow \frac{2m}{\pi} g R \left[ -\cos 90 - (-\cos 0) \right]$$

$$\Rightarrow \boxed{\frac{2m g R}{\pi}}$$

\* Spring Potential Energy.

$$U = \frac{1}{2} K x^2$$

• Work done by spring force

$$= \frac{1}{2} K (x_1^2 - x_2^2)$$



\* Potential Energy of Spring is zero at its natural length.

$$(*) \quad U_2 - U_1 = \frac{1}{2} K x_2^2 - \frac{1}{2} K x_1^2$$

- Either you have stretched  $x$ , or compressed  $x$  ~~far~~ from natural length the potential energy stored will always be equal to  $\left(\frac{1}{2} K x^2\right)$ .

• Potential Energy of spring is always positive.

- Gravitational & Electrostatic potential energy can be negative.

$$(*) \quad F_x = -\frac{du}{dx}$$

$$(*) \quad F_y = -\frac{du}{dy}$$

$$(*) \quad F_z = -\frac{du}{dz}$$



③ slope of  $U$  vs  $x$  graph is Conservative force.

## \* Equilibrium.

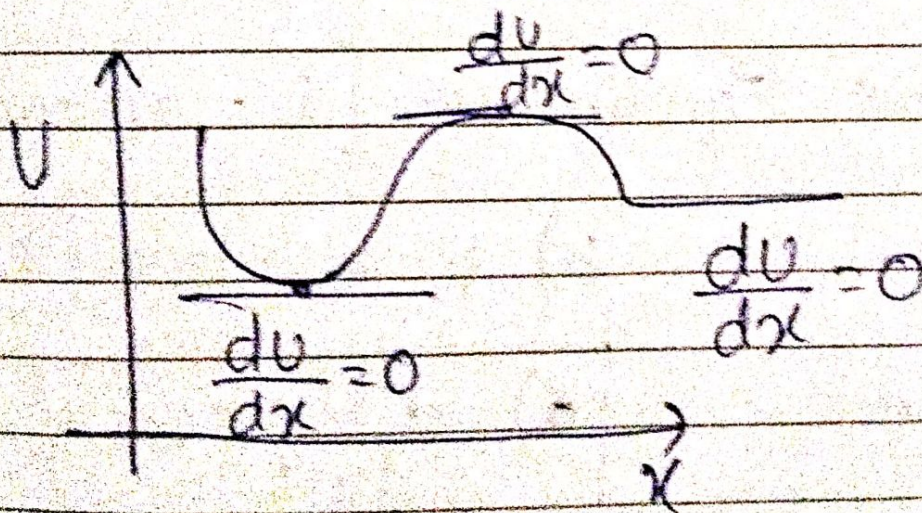
• If total force acting on body is zero, then body is in Equilibrium.

↓  
Stable

↓  
Unstable

↓  
Neutral

④  $F_x = -ve$  (slope  $U$  v/s  $x$ )



⑤ Stable Equilibrium  $\Rightarrow U \rightarrow$  minimum

⑥ Unstable Equilibrium  $\Rightarrow U \rightarrow$  maximum

⑦ Neutral Equilibrium  $\Rightarrow U \rightarrow$  constant.



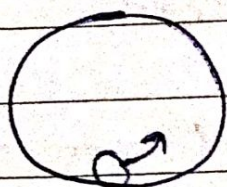
# 1. Stable Equilibrium [ $U = \text{minimum}$ ]

a.)  $F_{\text{net}} = 0$

b.)  $U-x$  slope  $= 0$

c.)  $\frac{dU}{dx} = 0$

d.)  $\frac{d^2U}{dx^2} > 0$



If we displace the body slightly it, returns back to stable equilibrium.

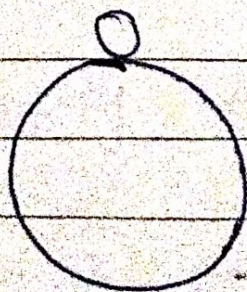
# 2. Unstable Equilibrium [ $U = \text{maximum}$ ]

a.)  $F_{\text{net}} = 0$

b.)  $U-x$  slope  $= 0$

c.)  $\frac{dU}{dx} = 0$

d.)  $\frac{d^2U}{dx^2} < 0$



If we displace the body slightly it will move towards ~~stable~~ equilibrium.  
Stable



### ③ Neutral Equilibrium [U-constant]

a)  $F_{\text{net}} = 0$

b)  $U-x$  Slope  $= 0$

c)  $\frac{dU}{dx} = 0$

d)  $\frac{d^2U}{dx^2} = 0$



will remain in again in neutral equilibrium position [new position].

### \* Conservation of Mechanical Energy

$$W_e + W_{N.C.} = (K_f + U_f) - (K_i + U_i)$$

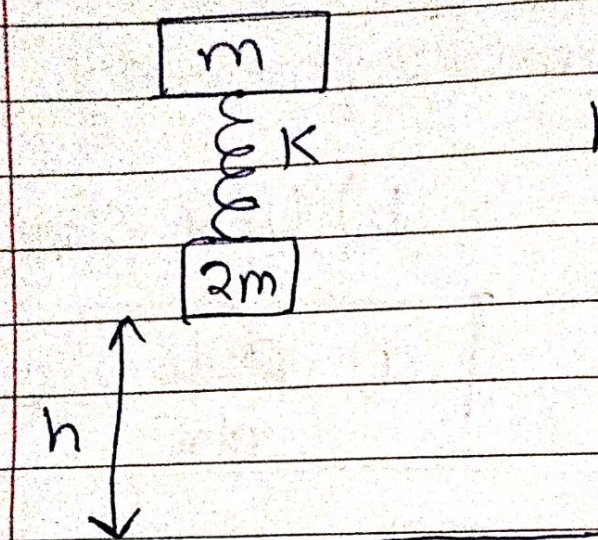
$\Rightarrow$  When external force  $= 0$   
Non-conservative force  $= 0$

then,  $K_i + U_i = K_f + U_f$

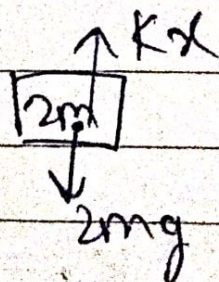
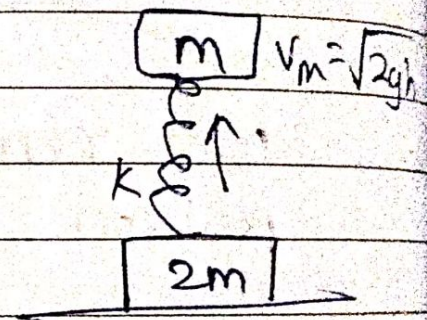
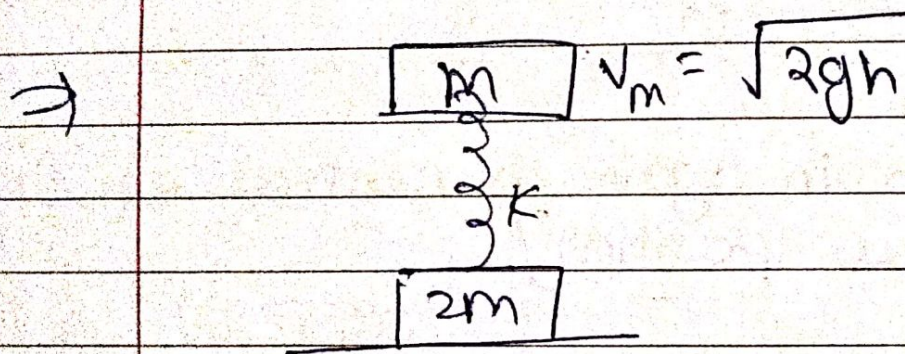


Ps  
ADV

Que

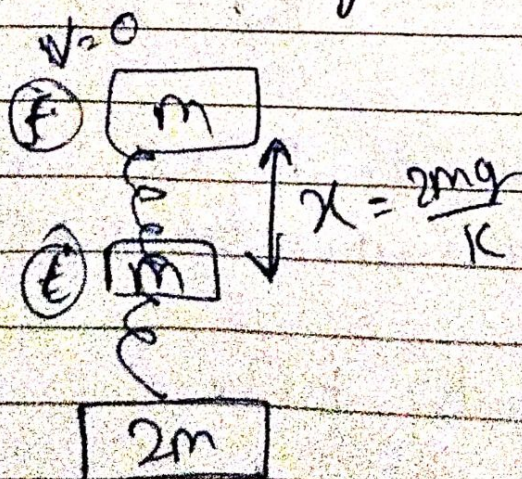


Find the value of  $h$  such that  $2m$  just manages to rise above ground after collision.



$Kx$  should just more  $\Rightarrow$  than  $2mg$

$\Rightarrow$  We can take  $Kx = 2mg$   
 $x = \frac{2mg}{K}$





Using Conservation of mechanical Energy

$$K_f + U_f = K_i + U_i$$

$$0 + mgx + \frac{1}{2}kx^2 = \frac{1}{2}m \cdot 2gh + 0$$

$$\Rightarrow mgx + \frac{1}{2}kx^2 = \frac{1 \cdot 2mgh}{2}$$

$$\Rightarrow \frac{2mgx + kx^2}{2} = mgh$$

$$\Rightarrow 2mgx + kx^2 = 2mgh$$

$$\Rightarrow kx^2 + 2mgx - 2mgh = 0$$

$$x = \frac{2mg}{k}$$

$$\Rightarrow k \cdot \frac{4m^2g^2}{k^2} + 2mg \frac{2mg}{k} - 2mgh = 0$$

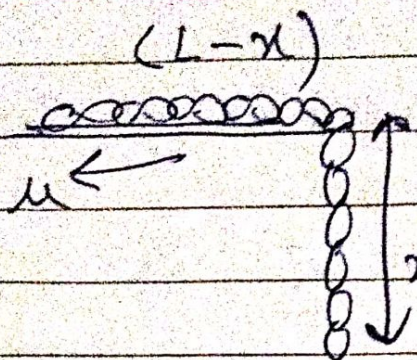
$$\Rightarrow \frac{4m^2g^2}{k} + \frac{4m^2g^2}{k} = 2mgh$$

$$\Rightarrow \frac{8m^2g^2}{2k} = mgh$$

$$\Rightarrow \frac{4m^2g^2}{k} = mgh \Rightarrow \boxed{h = \frac{4mg}{k}}$$



Que



A chain of length ' $L$ ' and mass ' $M$ ' is made to sustain on the table find the maximum length it can hang so that body doesn't fall.

$\Rightarrow$  Let the chain to be hanged be of length  $x$ .

$\therefore$  chain on table  $= (L-x)$

If the  $L$  length of chain weighs  $M$   
 $x$  length of chain weighs  $\frac{Mx}{L}$

As  $(L-x)$  length of chain weight  $\frac{M(L-x)}{L}$

Now frictional force  $= \mu N$

$$\Rightarrow \mu (mg)$$

$$\Rightarrow \mu \frac{M(L-x)}{L} g$$

Force acting on falling chain  $= \frac{Mx}{L} g$



→ A body has to stay

$$\therefore \mu \frac{M(L-x)g}{L} = \frac{Mxg}{L}$$

$$\Rightarrow \mu(L-x) = x$$

$$\Rightarrow \mu L - \mu x = x$$

$$\Rightarrow \mu L = x + \mu x$$

$$\Rightarrow \mu L = x(1+\mu)$$

$$\Rightarrow \boxed{\frac{\mu L}{1+\mu} = x}$$

Ques In the same Que above find the total work done by frictional force if the chain is pulled horizontal down.

→

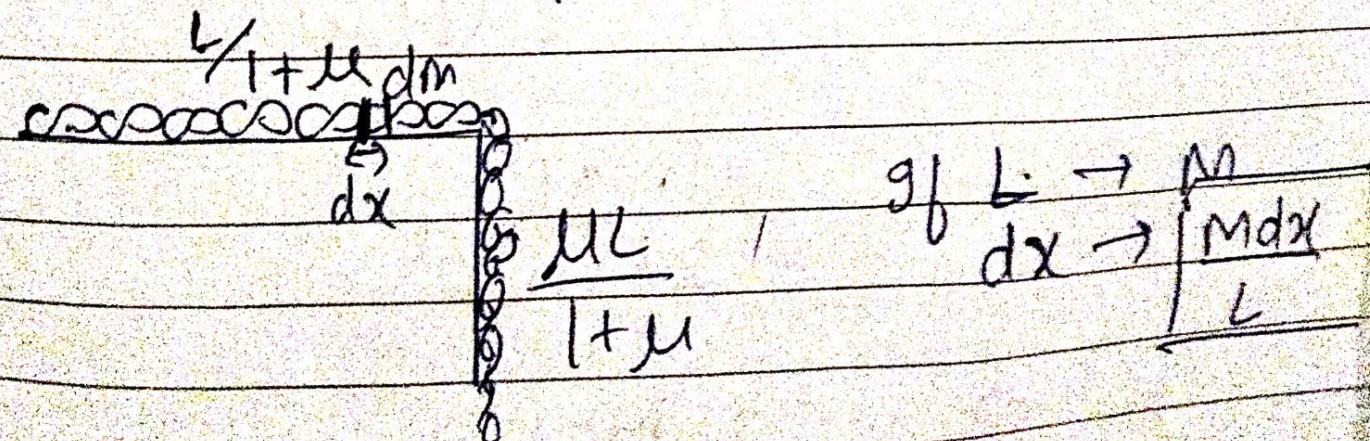


Diagram showing a chain of length  $L$  hanging vertically. A horizontal section of length  $x$  is being pulled down. The mass of the horizontal section is labeled as  $\frac{\mu}{1+\mu} dm$ . The mass of the vertical section is labeled as  $\frac{\mu L}{1+\mu}$ . To the right, a small diagram shows a horizontal line of length  $L$  with a mass  $M$ , and a vertical line of length  $L$  with a mass  $M$ , with a horizontal arrow pointing from the horizontal line to the vertical line.

$$f = \mu mg = \mu \frac{M dx}{L} g$$

$$dw = f dx$$



$$\int dw = \int \frac{\mu M dx}{L} g x$$

$$\Rightarrow \int dw = \frac{\mu M g}{L} \int x dx$$

vary  $x$  from  $\frac{L}{1+\mu}$  to 0.

$$\therefore \int dw = \frac{\mu M g}{L} \int_0^{\frac{L}{1+\mu}} x dx$$

$$\Rightarrow \frac{\mu M g}{L} \left[ \frac{x^2}{2} \right]_0^{\frac{L}{1+\mu}}$$

$$= \frac{\mu M g}{2L} \left[ \left( \frac{L}{1+\mu} \right)^2 \right]$$

$$\Rightarrow \frac{\mu M g L^2}{2k(1+\mu)^2}$$

$$\Rightarrow \frac{\mu M g L}{2(1+\mu)^2}$$